

TASK-2

Title: Synthesis and Gate-Level Simulation of 8-Bit FIFO using Vivado.

Objective: To synthesize an 8-bit FIFO design and verify its functionality through RTL and synthesized netlist analysis.

Tools Used: Vivado Simulator, Verilog HDL.

fifo.v

RTL code:

```
`timescale 1ns / 1ps

module fifo (
    input clk,
    input rst,
    input wr_en,
    input rd_en,
    input [7:0] data_in,
    output reg [7:0] data_out,
    output full,
    output empty
);

    reg [7:0] mem [0:7];
    reg [2:0] wr_ptr;
    reg [2:0] rd_ptr;
    reg [3:0] count;
```

```
assign full = (count == 8);
```

```
assign empty = (count == 0);
```

```
integer i;
```

```
always @(posedge clk or posedge rst)
```

```
begin
```

```
    if(rst)
```

```
    begin
```

```
        wr_ptr <= 0;
```

```
        rd_ptr <= 0;
```

```
        count <= 0;
```

```
        data_out <= 0;
```

```
        for(i=0;i<8;i=i+1)
```

```
            mem[i] <= 0;
```

```
    end
```

```
    else
```

```
    begin
```

```
        if(wr_en && !full)
```

```
        begin
```

```
        mem[wr_ptr] <= data_in;
        wr_ptr <= wr_ptr + 1;
        count <= count + 1;
    end

    if(rd_en && !empty)
    begin
        data_out <= mem[rd_ptr];
        rd_ptr <= rd_ptr + 1;
        count <= count - 1;
    end

end

end

endmodule
```

fifo_tb.v

Testbench code:

```
`timescale 1ns / 1ps

module fifo_tb;

    reg clk;

    reg rst;

    reg wr_en;
```

```
reg rd_en;
```

```
reg [7:0] data_in;
```

```
wire [7:0] data_out;
```

```
wire full;
```

```
wire empty;
```

```
fifo DUT(
```

```
    .clk(clk),
```

```
    .rst(rst),
```

```
    .wr_en(wr_en),
```

```
    .rd_en(rd_en),
```

```
    .data_in(data_in),
```

```
    .data_out(data_out),
```

```
    .full(full),
```

```
    .empty(empty)
```

```
);
```

```
always #5 clk = ~clk;
```

```
initial
```

```
begin
```

```
    clk = 0;
```

```
rst = 1;

wr_en = 0;

rd_en = 0;

data_in = 0;

#20;

rst = 0;

wr_en = 1;

data_in = 8'h11; #10;

data_in = 8'h22; #10;

data_in = 8'h33; #10;

data_in = 8'h44; #10;


wr_en = 0;


rd_en = 1;


#40;


rd_en = 0;

#20;

$finish;

end

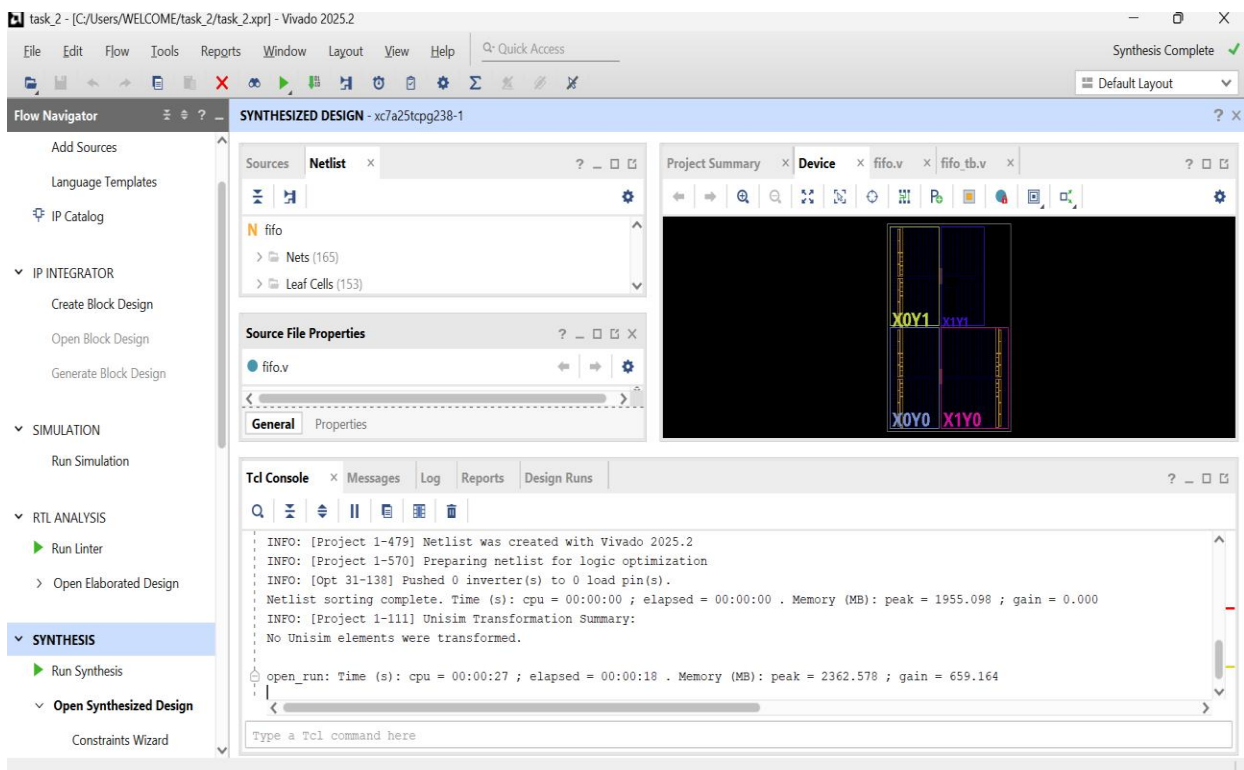
endmodule
```

RTL Simulation Results:

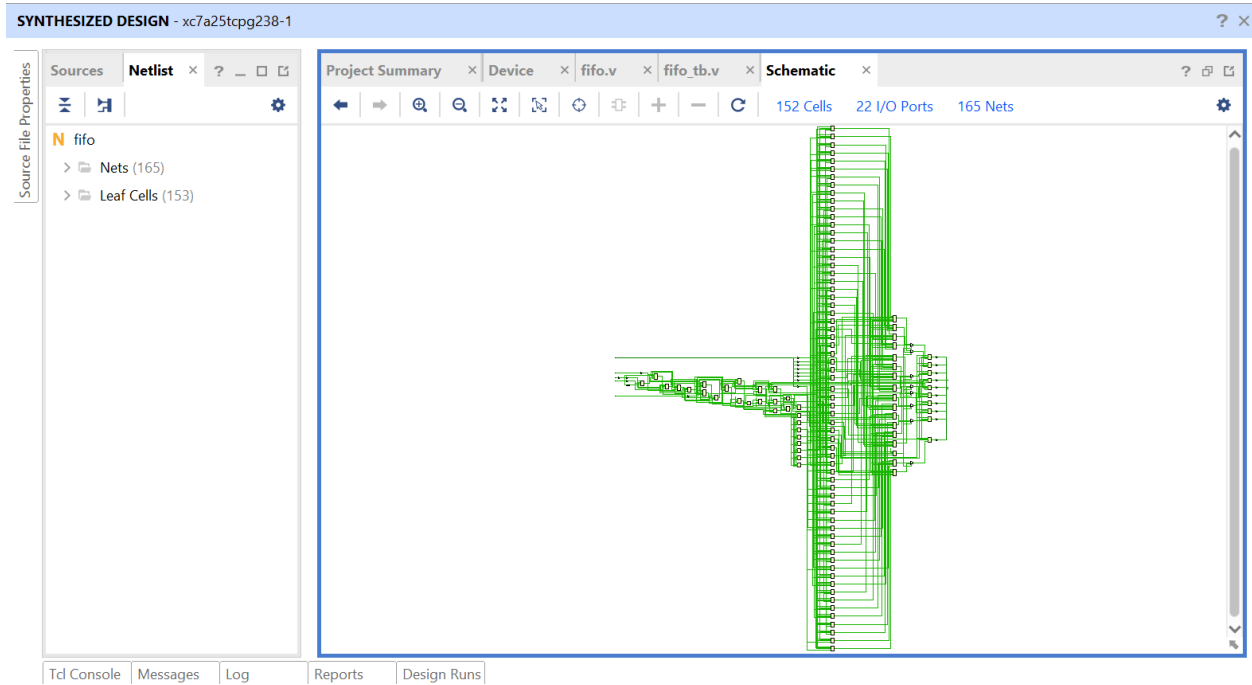


Synthesis Results:

Synthesis completion:



Schematic screenshot:



Utilization report screenshot:

SYNTHESIZED DESIGN - xc7a25tcbg238-1

Utilization ? _ □ ↻

Hierarchy

Name	Slice LUTs (14600)	Slice Registers (29200)	F7 Muxes (7300)	Bonded IOB (112)	BUFGCTRL (32)
N fifo	34	82	8	22	1

Source File Properties

Hierarchy

- Summary
- ▼ Slice Logic
 - ▼ Slice LUTs (<1%)
 - LUT as Logic (<1%)
 - ▼ Slice Registers (<1%)
 - Register as Flip Flop (<1%)
 - F7 Muxes (<1%)
- Memory
- DSP
- ▼ IO and GT Specific
 - Bonded IOB (20%)
- ▼ Clocking
 - BUFGCTRL (3%)
- Specific Feature
- Primitives
- Black Boxes
- Instantiated Netlists

utilization_1

Analysis:

The FIFO correctly stored and retrieved data values (11h, 22h, 33h, and 44h). The empty flag was asserted when the FIFO became empty and deasserted when data was available. Synthesis generated a gate-level implementation consisting of 152 cells and 165 nets.

Conclusion:

The FIFO design was successfully synthesized and verified. RTL simulation demonstrated correct FIFO operation, and the synthesized netlist and utilization report confirmed successful hardware implementation.